

NAAN MUDHALVAN

PROJECT DOCUMENTATION

**SALESFORCE CRM
IMPLEMENTATION**

Courier Management System

Platform	Salesforce Lightning Experience
Project Type	Courier Management System (CMS)
Purpose	Customer, account, order, automation, reporting and dashboard management

Submitted as part of the Naan Mudhalvan practical project documentation.

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1. INTRODUCTION

1.1 Project Overview

Courier businesses handle many activities such as customer registration, parcel booking, branch coordination, delivery assignment, status updates and billing. When these tasks are maintained through separate registers or spreadsheets, information can become difficult to locate and updates may be delayed.

The proposed Courier Management System (CMS) brings these activities into one Salesforce environment. It provides a structured way to store courier information, connect shipments with customers and delivery agents, automate selected processes, and present operational information through reports and dashboards.

The system is designed as a cloud-based academic project using Salesforce objects, relationships, validation rules, page layouts, flows, profiles, roles and reporting features.

1.2 Purpose

The purpose of CMS is to provide a centralized platform for recording and monitoring courier operations. The solution focuses on reducing repeated manual work, improving data consistency, supporting shipment visibility and helping staff review operational information.

Main objectives are:

- Maintain customer and shipment information in organized records.
- Connect shipments with delivery agents and branches.
- Use validation rules to reduce incorrect or incomplete entries.
- Automate selected status updates using Salesforce Flow.
- Provide role-based access for different users.
- Create reports and dashboards for operational monitoring.

1.3 User Story

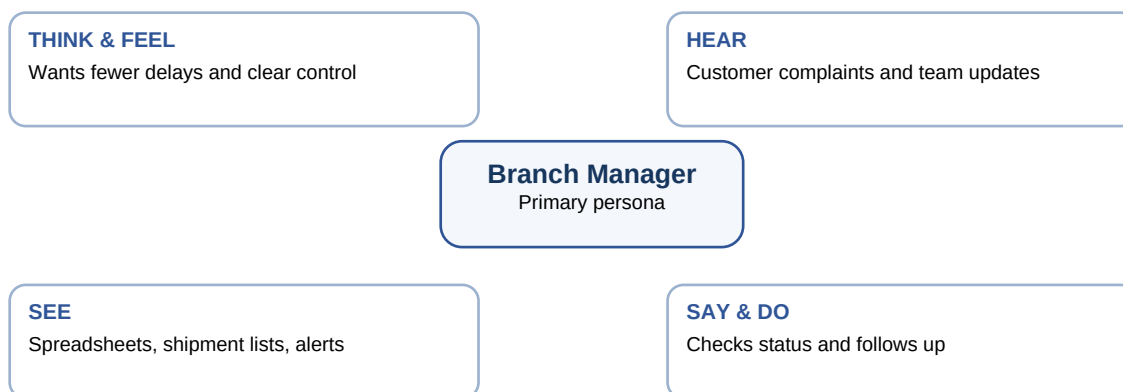
As a branch manager, I want to manage shipments, agents and delivery updates from one system so that daily courier operations can be monitored without depending on multiple manual records.

2. IDEATION PHASE

2.1 Problem Statement

User	Goal	Current Difficulty	Effect
Branch Manager	Manage shipments efficiently	Tracking is spread across records	More follow-up work
Customer	Know parcel status	Updates may not be centralized	Uncertainty about delivery
Delivery Agent	Receive clear assignments	Information may be shared manually	Risk of missed updates
Support Staff	Answer tracking queries	Need to check multiple records	Slower responses

2.2 Empathy Map



The branch manager is treated as the primary persona because this role needs visibility across shipments, agents and branch-level activities.

2.3 Brainstorming

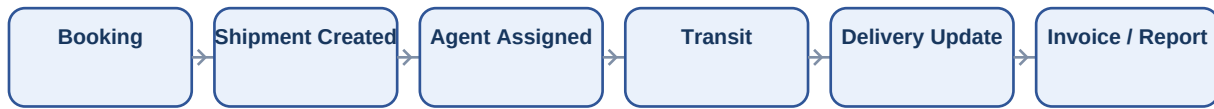
Idea	Strength	Limitation
Manual Register	Easy to start	Hard to search and update
Spreadsheet	Simple digital storage	Limited automation and coordination
Separate Web Portal	Can be customized	Requires additional infrastructure
Salesforce CRM	Cloud based with automation	Needs configuration and learning

2.4 Selected Approach

For this project, Salesforce CRM is used as the implementation platform because the reference project is built around Salesforce features such as custom objects, Flow automation, security settings and reports. The student implementation can therefore demonstrate an end-to-end business application without building every infrastructure component from scratch.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey



The journey begins when shipment information is recorded and continues through assignment, movement, delivery status updates and business reporting.

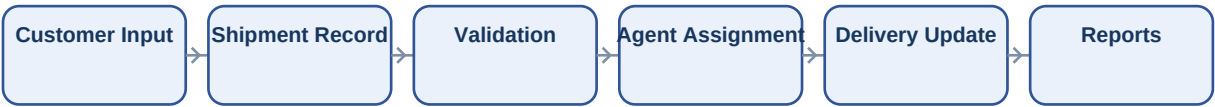
3.2 Functional Requirements

ID	Requirement
FR-01	Create and maintain customer records.
FR-02	Create shipment records with source, destination, weight, price and status.
FR-03	Associate a shipment with a delivery agent.
FR-04	Record delivery status updates.
FR-05	Maintain branch and invoice information.
FR-06	Generate operational reports and dashboards.
FR-07	Apply validation rules to important fields.

3.3 Non-Functional Requirements

Area	Requirement
Usability	A clear Salesforce Lightning interface should be used.
Security	Access should be separated according to user responsibilities.
Reliability	Automation should execute consistently for valid records.
Performance	Common records and tracking information should load efficiently.
Availability	The cloud platform should support access when required.
Scalability	The design should allow additional branches and records.

3.4 Data Flow



3.5 Technology Stack

Layer	Salesforce Feature
User Interface	Salesforce Lightning
Business Logic	Salesforce Flow
Data Storage	Custom Salesforce Objects
Data Validation	Validation Rules
Security	Profiles, Roles and Permissions
Analytics	Reports and Dashboards

4. PROJECT DESIGN

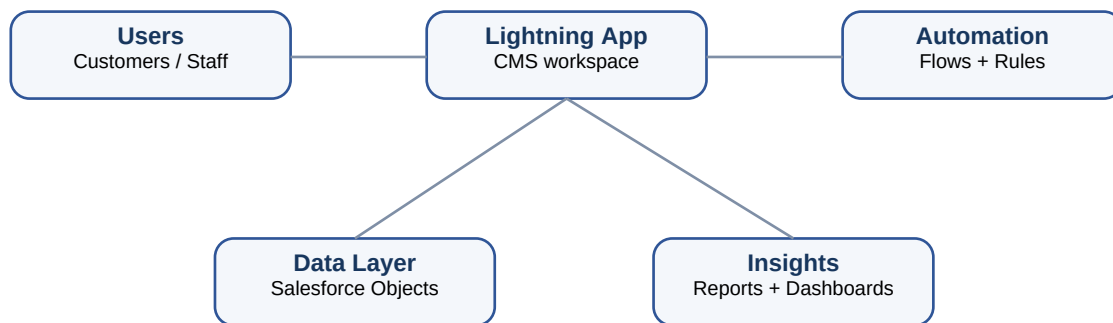
4.1 Problem–Solution Fit

Business Issue	CMS Response
Scattered shipment information	Centralized Shipment records
Incorrect data entries	Validation rules
Manual status propagation	Record-triggered Flow
Unclear user permissions	Profiles and roles
Difficult performance review	Reports and dashboards

4.2 Proposed Solution

CMS is organized around connected Salesforce objects. Customer information is linked to shipments; shipments can reference delivery agents; agents are associated with branches; delivery updates belong to shipments; and invoices can reference shipment records.

4.3 Solution Architecture



4.4 Main Components

- Customer – stores customer identity and contact details.
- Shipment – stores tracking, address, weight, price, status and expected date.
- Delivery Agent – stores agent contact, branch and working status.
- Branch – represents courier office information.
- Delivery Update – stores shipment status changes and remarks.
- Invoice – stores billing and payment information.
- Flows – automate selected record updates.
- Reports and Dashboards – present operational information.

5. PROJECT PLANNING & SCHEDULING

5.1 Planning Approach

The project follows a simple Agile-style approach in which the work is divided into manageable sprints. Each sprint focuses on a functional area of the application.

Sprint	Epic	Example User Story	Points	Priority
1	Platform Setup	Create and configure the Salesforce project environment.	3	High
2	Data Modeling	Create the core CMS objects and relationships.	5	High
3	Automation	Add validations and shipment status automation.	5	High
4	Security	Configure profiles and role hierarchy.	5	High
5	Reporting	Build operational reports.	4	High
6	Dashboard	Create dashboard components for monitoring.	4	Medium

5.2 Project Tracker

Sprint	Planned Focus	Expected Output
Sprint 1	Environment	Developer org and initial configuration
Sprint 2	Objects	Customer, Shipment, Agent, Branch, Update and Invoice
Sprint 3	Rules & Flow	Validation rules and status automation
Sprint 4	Security	Profiles, roles and user access
Sprint 5	Analytics	Reports
Sprint 6	Visualization	Dashboard

5.3 Estimation Note

Story points are used as relative estimates. A larger point value represents work that generally needs more configuration or testing. The exact duration can be adjusted according to the student team's schedule and Salesforce environment.

6. PROJECT DEVELOPMENT PHASE

6.1 Salesforce Developer Account

Salesforce provides a cloud environment in which the CMS configuration can be created. A developer account is used for learning and project development.

Typical setup sequence:

- Open the Salesforce Developer sign-up page.
- Enter the requested personal and organization information.
- Activate the account through the verification email.
- Set the password and complete the initial login.
- Open Setup to begin configuring the application.

6.2 Understanding Salesforce Objects

A Salesforce object can be understood as a structured data container. Fields store individual values and records represent individual entries.

Object	Purpose
Customer__c	Customer profile and contact information
Shipment__c	Parcel and tracking information
Delivery_Agent__c	Delivery personnel details
Branch__c	Courier branch information
Delivery_Update__c	Shipment status history
Invoice__c	Billing and payment information

6.3 Object Relationships

Child	Field	Parent	Relationship
Shipment	Customer__c	Customer	Lookup
Shipment	Assigned_Agent__c	Delivery Agent	Lookup
Delivery Agent	Branch__c	Branch	Lookup
Delivery Update	Shipment__c	Shipment	Master-Detail
Invoice	Shipment__c	Shipment	Lookup

6.4 Customer Fields

Field	API Name	Type	Required
Customer Name	Name	Text	Yes
Email	Email__c	Email	Yes
Phone	Phone__c	Phone	Yes
Address	Address__c	Text Area	Yes
Customer Type	Customer_Type__c	Picklist	Yes
City	City__c	Text	No
State	State__c	Text	No
Pincode	Pincode__c	Text	No
Is Active	Is_Active__c	Checkbox	No

6.5 Shipment Fields

Field	API Name	Type	Required
Tracking Number	Name	Auto Number	Yes
Customer	Customer__c	Lookup	Yes
Source Address	Source_Address__c	Text Area	Yes
Destination Address	Destination_Address__c	Text Area	Yes
Weight	Weight__c	Number	Yes
Price	Price__c	Currency	Yes
Status	Status__c	Picklist	Yes
Expected Date	Expected_Date__c	Date	Yes
Assigned Agent	Assigned_Agent__c	Lookup	No
Shipment Type	Shipment_Type__c	Picklist	Yes

6.6 Delivery Agent, Branch, Update and Invoice Fields

Object	Important Fields
Delivery Agent	Agent Name, Email, Phone, Branch, Status, Joining Date
Branch	Branch Name, Branch Code, City, State, Manager Name
Delivery Update	Update Number, Shipment, Status, Update Date, Remarks
Invoice	Invoice Number, Shipment, Amount, Invoice Date, Payment Status

6.7 Creating Custom Objects – Activity Guide

For each custom object, open Setup → Object Manager → Create → Custom Object. Enter the label, plural label and record name type. Enable the required optional features and save the object.

Object	Suggested Record Name
Customer	Customer Name (Text)
Shipment	Tracking Number (Auto Number)
Delivery Agent	Agent Name (Text)
Branch	Branch Name (Text)
Delivery Update	Update Number (Auto Number)
Invoice	Invoice Number (Auto Number)

6.8 Creating Tabs

Custom tabs make the objects easier to access from the Lightning application. Create tabs for Customer, Shipment, Delivery Agent, Branch, Delivery Update and Invoice, selecting an appropriate tab style for each.

6.9 Creating Fields

For each object, use Fields & Relationships → New, choose the required data type, provide the field label, configure required settings or picklist values, and save. Relationship fields should be created using Lookup or Master-Detail according to the data model.

6.10 Validation Rules

Validation rules check whether entered information meets a defined condition. When the condition evaluates as true, Salesforce displays an error and prevents the record from being saved.

Object	Rule	Formula / Condition	Message
Customer	Email required	ISBLANK(Email__c)	Email is required.
Customer	Phone length	LEN(Phone__c) <> 10	Phone number must be 10 digits.
Customer	Pincode length	LEN(Pincode__c) <> 6	Pincode must be 6 digits.
Shipment	Positive weight	Weight__c <= 0	Weight must be greater than zero.
Shipment	Positive price	Price__c <= 0	Price must be greater than zero.
Shipment	Future expected date	Expected_Date__c < TODAY()	Expected delivery date cannot be in the past.
Delivery Agent	Email format check	NOT(CONTAINS(Email__c, "@"))	Enter a valid email address.
Branch	Branch code required	ISBLANK(Branch_Code__c)	Branch Code is mandatory.

6.11 Delayed Shipment Remark Rule

A delivery update marked as Delayed should contain a remark. A suitable condition is:

AND(ISPICKVAL(Status__c, "Delayed"), ISBLANK(Remarks__c))

Error message: **Remarks are required when status is Delayed.**

6.12 Page Layouts

Page layouts control how fields and related information are arranged on a Salesforce record page. The CMS can group fields into sections so users can enter information in a logical order.

Object	Suggested Sections
Customer	Customer Information; Address Details
Shipment	Shipment Details; Address Information; Delivery Information
Delivery Agent	Agent Details; Additional Information
Branch	Branch Information
Delivery Update	Update Information
Invoice	Invoice Details

6.13 Record-Triggered Flow

A record-triggered Flow can update the shipment status whenever a new Delivery Update record is created.



Configuration summary:

- Create a new Record-Triggered Flow.
- Object: Delivery_Update__c.
- Trigger: record is created.
- Use Update Records to find Shipment__c where Id equals the Delivery Update's Shipment__c.
- Set Shipment__c.Status__c to the Delivery Update Status__c.
- Save the flow and activate it.

6.14 Lightning Application

Create a Lightning App named **Courier Management System**. Add the main CMS tabs so staff can move between customers, shipments, agents, branches, delivery updates, invoices, reports and dashboards.

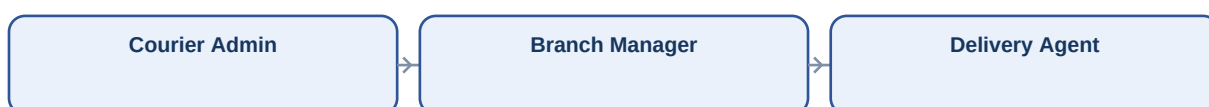
Suggested navigation items:

- Customers
- Shipments
- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

6.15 Profiles

Profile	Main Purpose
Courier Admin	Full configuration and administrative control
Branch Manager	Manage branch-level shipment and agent operations
Delivery Agent	Update assigned shipment delivery information
Customer Support	Handle customer and tracking enquiries

6.16 Role Hierarchy



Customer Support can be positioned under the Branch Manager when a separate support role is required. The hierarchy is used to organize reporting relationships and data visibility.

6.17 Users

A Salesforce user is an individual account that can log in to the organization. User access is determined by settings such as license, profile and role.

User Type	Role	Profile
Courier Admin	Courier Admin	Courier Admin
Branch Manager	Branch Manager	Branch Manager
Delivery Agent	Delivery Agent	Delivery Agent
Customer Support	Customer Support	Customer Support

When creating a user, provide a unique username, valid email, role and profile. The available licenses depend on the Salesforce environment.

6.18 Reports

Reports convert stored records into useful summaries. The following reports are suitable for the CMS:

Report	Main Fields / Grouping
Shipment Status Summary	Tracking Number, Customer, Status, Expected Date, Assigned Agent
Agent Performance	Agent Name, Branch, Shipment Count, Status
Revenue by Branch	Tracking Number, Customer, Price, Invoice Status, Branch, Month
Customer Summary	Customer details and related shipment information

6.19 Dashboard

A CMS dashboard can contain several components that provide a quick visual overview.

Component	Source	Visualization
Shipment Status Overview	Shipment Status Summary	Donut Chart
Agent Performance	Agent Performance Report	Bar Chart
Revenue by Branch	Revenue Report	Column Chart

Recommended dashboard name: **Courier Management Dashboard**.

7. FUNCTIONAL & PERFORMANCE TESTING

Testing is used to confirm that records, validation rules, automation, access settings and reports behave as intended.

Test ID	Scenario	Expected Result
T01	Create a valid customer	Customer record saves successfully
T02	Enter an invalid phone length	Validation message appears
T03	Enter a non-six-digit pincode	Pincode validation prevents save
T04	Enter zero shipment weight	Weight validation prevents save
T05	Enter a past expected date	Date validation prevents save
T06	Create a Delivery Update	Related shipment status is updated by Flow
T07	Create report	Correct shipment data is displayed
T08	Open dashboard	Charts display the configured report data
T09	Login as restricted user	Only permitted objects/actions are available

7.1 Performance Review

For this academic project, performance can be reviewed by checking whether common records open correctly, Flow execution completes without errors, and reports return the expected records. Large-scale performance measurements are not established by the reference document and should only be reported if the project team actually performs them.

8. ADVANTAGES & LIMITATIONS

8.1 Advantages

- Centralized storage of courier information.
- Reduced dependence on repeated manual status updates.
- Validation helps prevent selected data-entry errors.
- Flow automation can synchronize shipment status.
- Reports and dashboards provide operational visibility.
- Profiles and roles support controlled access.

8.2 Limitations

- The solution depends on Salesforce configuration and availability.
- Initial setup requires time and platform knowledge.
- Actual SMS, GPS and mobile features require additional implementation.
- Permissions and licenses must be configured correctly for users.

8.3 Risk Considerations

Risk	Possible Response
Incorrect configuration	Test each object, rule and flow before final use
Duplicate or incomplete data	Use validation and clear field requirements
Unauthorized access	Review profiles, roles and permissions
Automation error	Use Flow debugging and controlled test records
Report mismatch	Check filters, fields and source report types

9. CONCLUSION

The Courier Management System demonstrates how Salesforce can be configured to support courier operations through structured data, relationships, validation, automation, security and analytics. The design connects customers, shipments, delivery agents, branches, updates and invoices into a common application.

The project also demonstrates the complete development cycle from problem identification and requirements to design, configuration, testing and reporting. The resulting system provides a practical academic example of a cloud-based business application.

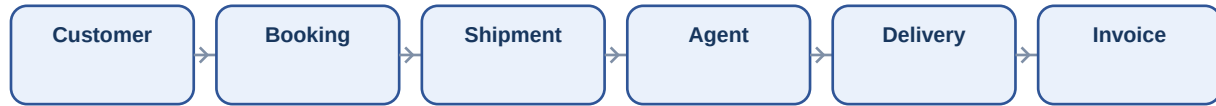
10. FUTURE SCOPE

- Mobile-friendly shipment tracking application.
- GPS-based delivery location integration.
- SMS or email notifications for shipment events.
- Additional branch-level analytics.
- AI-assisted delivery route or workload optimization.
- Customer self-service tracking interface.

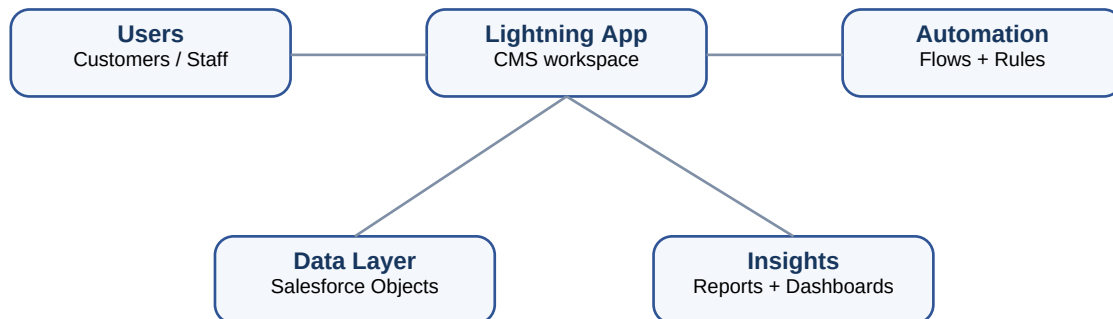
Future features should be treated as proposed extensions; they are not claimed as implemented in this document.

11. APPENDIX – PROJECT VISUALS

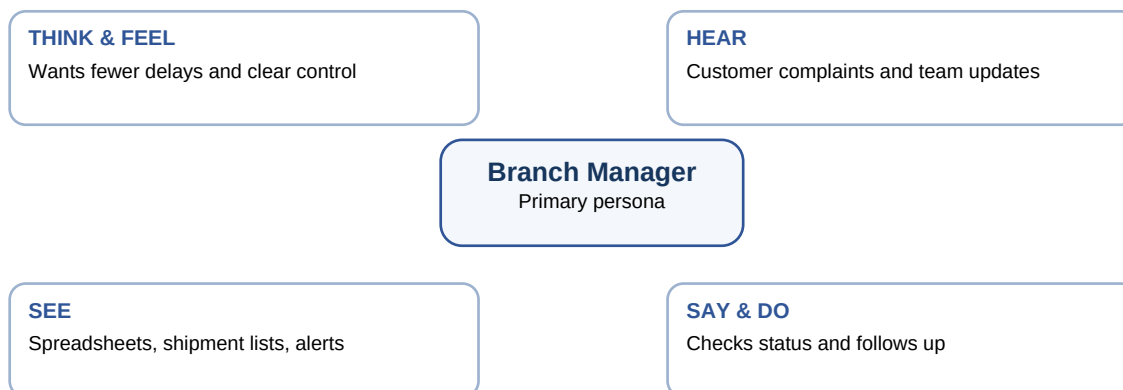
11.1 End-to-End Process Visual



11.2 CMS Architecture Visual



11.3 User Persona Visual



11.4 Data Relationship Summary

Customer	→	Shipment	One customer can be linked with shipment records
Shipment	→	Delivery Agent	A shipment can be assigned to an agent
Delivery Agent	→	Branch	Agent belongs to a branch
Shipment	→	Delivery Update	Updates record shipment progress
Shipment	→	Invoice	Billing record references shipment

11.5 Suggested Salesforce Screenshots for Final Submission

The following screenshots can be inserted from your own Salesforce org after completing the configuration. Using your own screenshots will make the submission clearly reflect your implementation.

- Salesforce Setup / Object Manager showing CMS objects.
- Customer object fields and a sample customer record.
- Shipment record showing tracking, status and assigned agent.
- Validation Rule screen for Weight or Pincode.
- Flow Builder screen showing the shipment status automation.
- Lightning App navigation showing CMS modules.
- Profile or role configuration.
- Shipment Status report.
- Agent Performance report.
- Courier Management Dashboard.

11.6 Student Submission Note

Prepared by: **YUSRA MEHEK N**

Course: BCA – Third Year

Project: Courier Management System (CMS)

Platform: Salesforce

THANK YOU